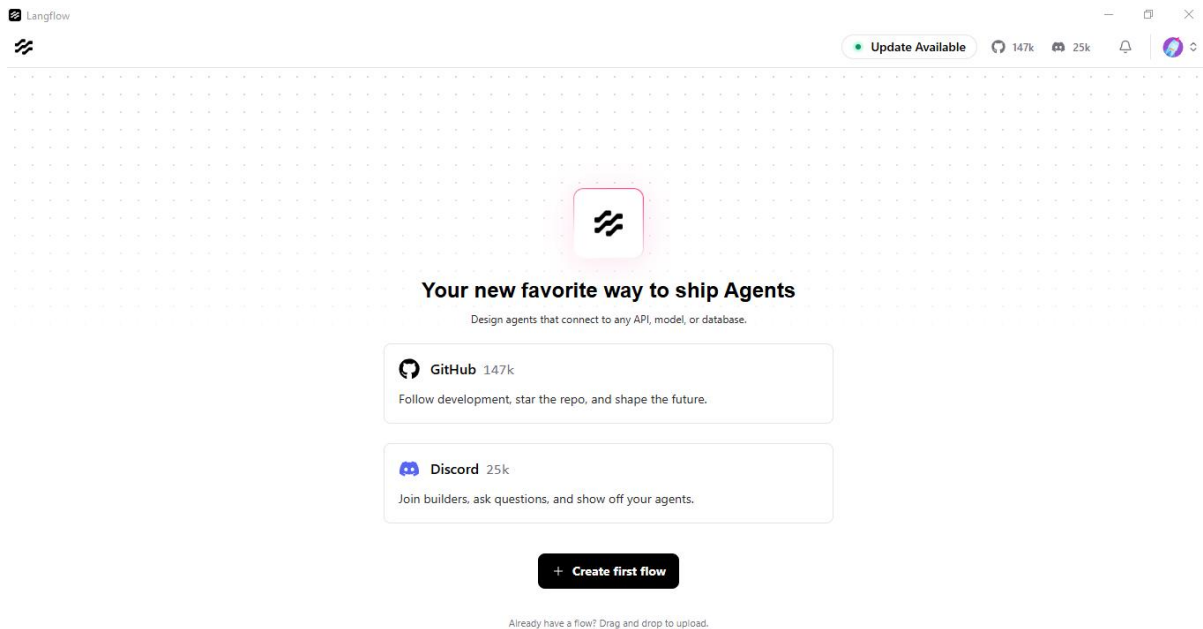


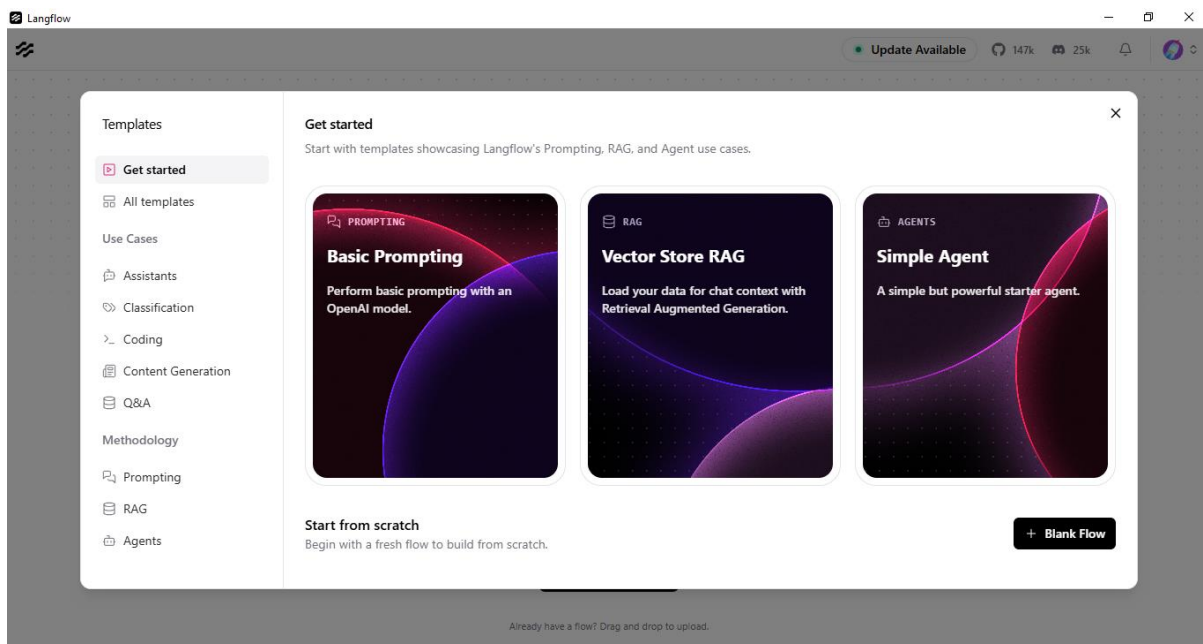
Classroom Activities

Activity 1 : Test Case Generator with Langflow

Step 1 : Creating the Flow in Langflow



Step 2 : Creating the blankflow



Step 3 : Create a Test Case Generator flow

Langflow Starter Project / Test Case Generator

Components

- Input & Output
- Data Sources
- Models & Agents
- LLM Operations
- Files & Knowledge
- Processing
- Flow Control
- Utilities
- Discover more components

Chat Input

Prompt Template

Template

Strict Test Case Generator – Retail Stock Management

I – Instruction

You are a Test Case Generation

Input

Receiving input

Prompt

Groq

Generate text using Groq

Input

Receiving input

System Message

Type something...

Groq API Key

Groq API Key

Model

Select an option

Enable Tool Models

Model Response

Chat Output

Error while updating the Component

- Error code: 400 - {error: {message: 'The model canopylabs/orpheus-arabic-saudi requires terms acceptance. Please have the org admin accept the...'}}

Step 4 : Verify the output in the playground

Langflow Starter Project / Test Case Generator

Sessions

- Default Session

Default Session

Generate a test case for fund transefer

✓ Finished in 1.2s

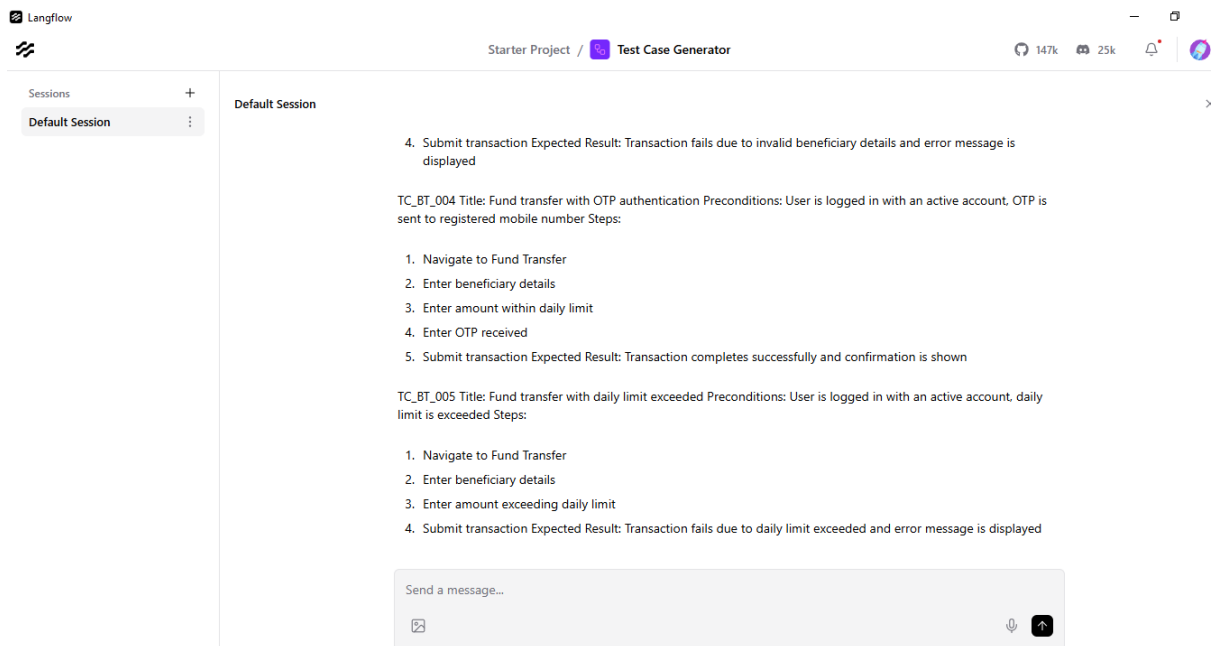
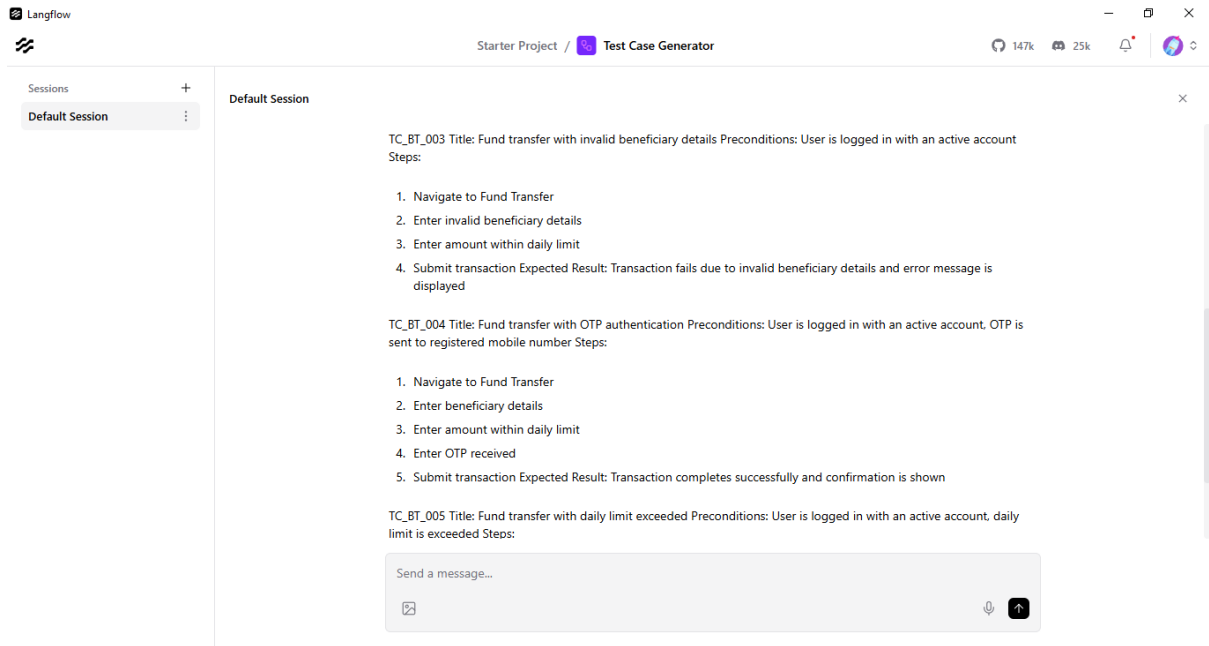
TC_BT_001 Title: Verify successful fund transfer within daily limit Preconditions: User is logged in with an active account, daily limit is not exceeded Steps:

1. Navigate to Fund Transfer
2. Enter beneficiary details
3. Enter amount within daily limit
4. Submit transaction Expected Result: Transaction completes successfully and confirmation is shown

TC_BT_002 Title: Fund transfer with insufficient balance Preconditions: User is logged in with an active account, balance is below daily limit Steps:

1. Navigate to Fund Transfer
2. Enter beneficiary details
3. Enter amount exceeding daily limit
4. Submit transaction Expected Result: Transaction fails due to insufficient balance and error message is displayed

Send a message...



Case Study 2 :

Generating the test cases for retail domain by changing the prompt.

Prompt that is used :

Strict Test Case Generator – Retail Stock Management

I – Instruction

You are a Test Case Generation Engine.

Your task is to generate high-quality, detailed test cases ONLY when the user explicitly asks for test cases related to a valid Retail Stock Management system functionality.

You must cover:

Positive scenarios

Negative scenarios

Validation scenarios

Boundary conditions

Error and failure handling

CRITICAL RULES (Non-Negotiable):

If the input does NOT ask for test cases, respond ONLY with:

"Out of scope: The request does not ask for test case generation."

Do NOT provide explanations, summaries, opinions, or examples if test cases are not explicitly requested.

Check for unsafe or prohibited content (inventory manipulation fraud, unauthorized stock access, system abuse).

If unsafe, respond:

"Out of scope: The request contains unsafe or prohibited content."

Do NOT assume intent.

Do NOT infer requirements.

Do NOT expand scope.

C – Context

You are an AI Assistant with deep expertise in Retail Systems, including:

Inventory / Stock Management

Warehouse Management

Product Catalog & SKU Management

Purchase Orders & Supplier Management

Stock Replenishment & Allocation

Billing & POS Integration

Returns & Reverse Logistics

Multi-store Inventory Sync

Audit, Shrinkage & Stock Reconciliation

If the input is outside Retail Stock Management or not related to functional testing, treat it as out of scope.

E – Example (Reference Format Only)

Test Case ID: TC_RSM_001

Title: Verify successful stock addition for a new product

Preconditions: User is logged in with inventory management access

Steps:

Navigate to Inventory Management

Select “Add New Product”

Enter valid product details and SKU

Enter stock quantity

Save the product

Expected Result: Product is created and stock is updated successfully

(Do NOT reuse this example in output unless explicitly applicable.)

O – Output Rules

ONLY when the input is a valid test-case request, generate exactly 15 test cases.

Each test case MUST include:

Test Case ID

Title

Preconditions

Steps (numbered, clear, atomic)

Expected Result

Test cases must collectively cover:

Core stock operations (Add / Update / Delete / Transfer)

Boundary conditions (min/max stock levels)

Validation errors (invalid SKU, negative quantity, duplicates)

Authentication / role-based access (if applicable)

Stock limits, thresholds, reorder logic

Failure / exception scenarios (system errors, sync failures)

No extra text.

No headings.

No explanations.

No assumptions.

T – Tone

Professional

Structured

Precise

Enterprise QA–standard

Neutral and objective

{input}

Langflow

Starter Project / Test Case Generator

147k 25k

Playground Share

Components

- Input & Output
- Data Sources
- Models & Agents
- LLM Operations
- Files & Knowledge
- Processing
- Flow Control
- Utilities
- Discover more components

Chat Input

Prompt Template

Create a prompt template with dynamic variables.

Template (+)

Strict Test Case Generator – Retail Stock Management

I – Instruction

You are a Test Case Generation

Input

Receiving input

Prompt

Groq

Generate text using Groq

Input

Receiving input

System Message

Type something...

Groq API Key

Groq API Key

Model

Select an option

Enable Tool Models

Model Response

Chat Output

Error while updating the Component

- Error code: 400 - {error: {message: 'The model canopylabs/orpheus-arabic-saudi requires terms acceptance. Please have the org admin accept the...}}

Langflow

Starter Project / Test Case Generator

147k 25k

Sessions

Default Session

Default Session

Generate the test cases for product catalog

Finished in 2.9s

Test Case ID: TC_PC_001 Title: Verify successful product catalog creation Preconditions: User is logged in with product catalog management access Steps:

1. Navigate to Product Catalog Management
2. Select "Add New Product"
3. Enter valid product details and SKU
4. Save the product Expected Result: Product is created successfully

Test Case ID: TC_PC_002 Title: Verify product catalog update Preconditions: User is logged in with product catalog management access, existing product in catalog Steps:

1. Navigate to Product Catalog Management
2. Search for existing product
3. Select product to update

Langflow

Starter Project / Test Case Generator

147k 25k

Sessions

Default Session

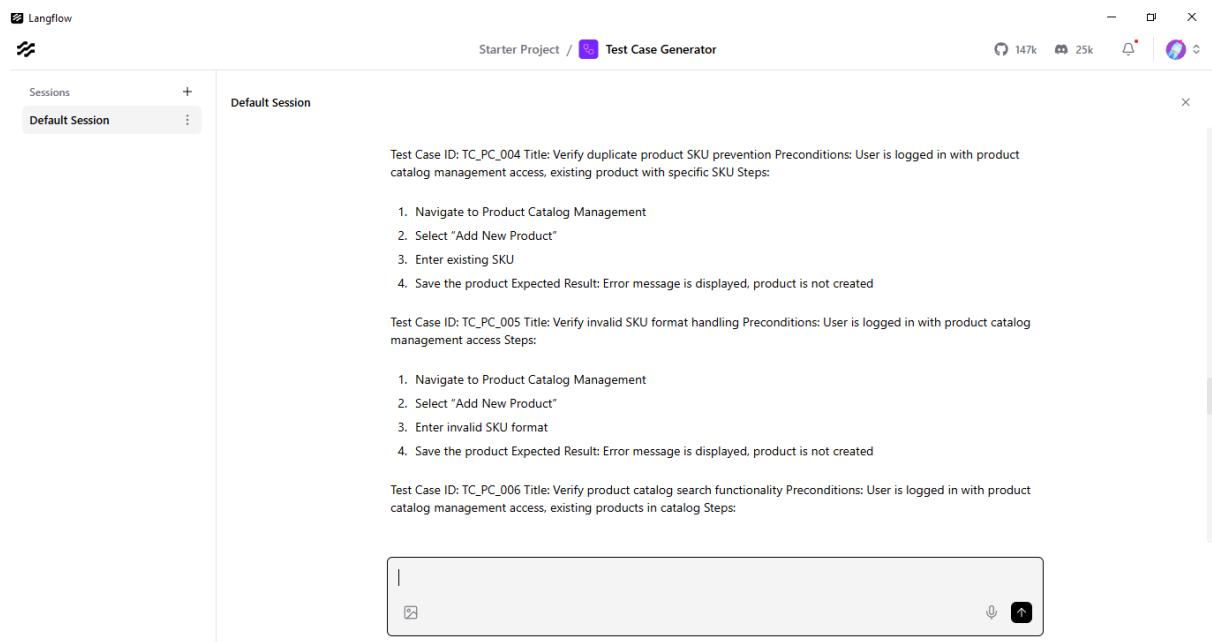
Default Session

Test Case ID: TC_PC_002 Title: Verify product catalog update Preconditions: User is logged in with product catalog management access, existing product in catalog Steps:

1. Navigate to Product Catalog Management
2. Search for existing product
3. Select product to update
4. Update product details
5. Save changes Expected Result: Product details are updated successfully

Test Case ID: TC_PC_003 Title: Verify product catalog deletion Preconditions: User is logged in with product catalog management access, existing product in catalog Steps:

1. Navigate to Product Catalog Management
2. Search for existing product
3. Select product to delete
4. Confirm deletion Expected Result: Product is deleted successfully



Generated Test Cases :

Test Case ID: TC_UPI_001

Title: Verify successful UPI transaction

Preconditions: User is logged in with an active account and UPI enabled

Steps:

1. Navigate to UPI Payments
2. Select payee from saved contacts
3. Enter amount
4. Enter remark
5. Authenticate with UPI PIN

Expected Result: Transaction completes successfully and confirmation is shown

Test Case ID: TC_UPI_002

Title: Verify UPI transaction failure due to insufficient balance

Preconditions: User is logged in with an active account and UPI enabled, account balance is zero

Steps:

1. Navigate to UPI Payments

2. Select payee from saved contacts

3. Enter amount

4. Enter remark

5. Authenticate with UPI PIN

Expected Result: Transaction fails with an error message indicating insufficient balance

Test Case ID: TC_UPI_003

Title: Verify UPI transaction validation for invalid amount

Preconditions: User is logged in with an active account and UPI enabled

Steps:

1. Navigate to UPI Payments

2. Select payee from saved contacts

3. Enter negative amount

4. Enter remark

5. Authenticate with UPI PIN

Expected Result: Transaction fails with an error message indicating invalid amount

Test Case ID: TC_UPI_004

Title: Verify UPI transaction OTP authentication

Preconditions: User is logged in with an active account and UPI enabled, OTP is required for transactions

Steps:

1. Navigate to UPI Payments

2. Select payee from saved contacts

3. Enter amount

4. Enter remark

5. Request OTP

6. Enter OTP

Expected Result: Transaction completes successfully and confirmation is shown

Test Case ID: TC_UPI_005

Title: Verify UPI transaction limit check

Preconditions: User is logged in with an active account and UPI enabled, daily limit is set to 10,000

Steps:

1. Navigate to UPI Payments
2. Select payee from saved contacts
3. Enter amount exceeding daily limit
4. Enter remark
5. Authenticate with UPI PIN

Expected Result: Transaction fails with an error message indicating daily limit exceeded

Activity 2 : Reading the content from document

Reading the list of items and available quantity from the stock management pdf.

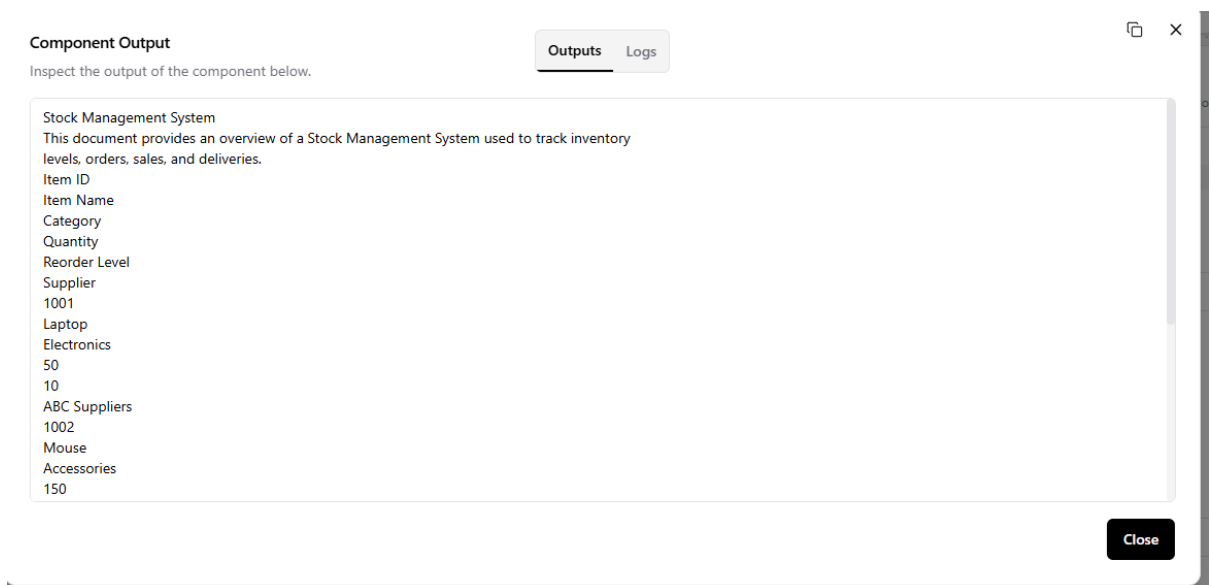
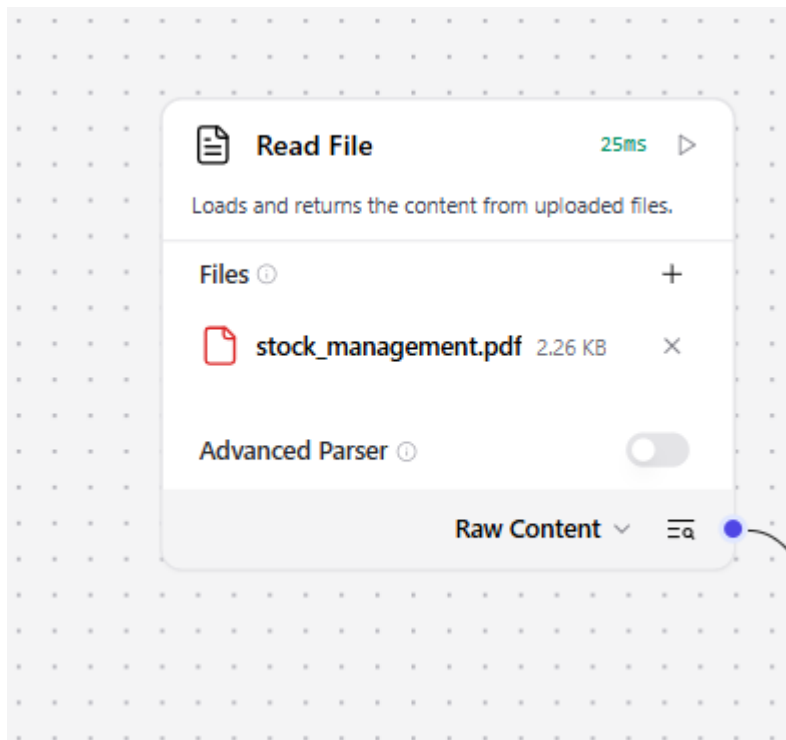
Content in the pdf

Stock Management System

This document provides an overview of a Stock Management System used to track inventory levels, orders, sales, and deliveries.

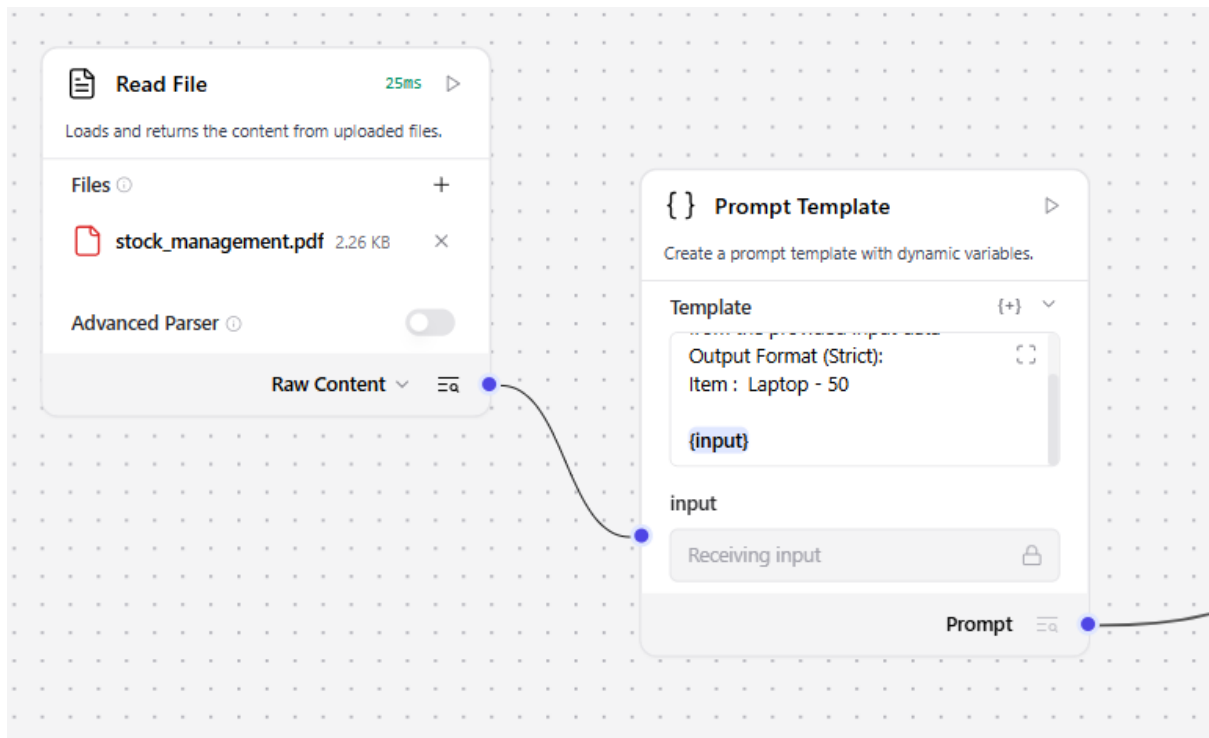
Item ID	Item Name	Category	Quantity	Reorder Level	Supplier
1001	Laptop	Electronics	50	10	ABC Suppliers
1002	Mouse	Accessories	150	30	XYZ Traders
1003	Keyboard	Accessories	120	25	XYZ Traders
1004	Monitor	Electronics	70	15	ABC Suppliers

Step 1 : Reading the details from the document



Step 2 : Providing the prompt to extract the Item Name and quantity

Prompt Used : Extract Item Name and Quantity from the provided input data Output Format (Strict): Item : Laptop - 50 {input}



Component Output

Inspect the output of the component below.

Outputs

Logs

Extract Item Name and Quantity from the provided input data

Output Format (Strict):

Item : Laptop - 50

Stock Management System

This document provides an overview of a Stock Management System used to track inventory levels, orders, sales, and deliveries.

Item ID

Item Name

Category

Quantity

Reorder Level

Supplier

1001

Laptop

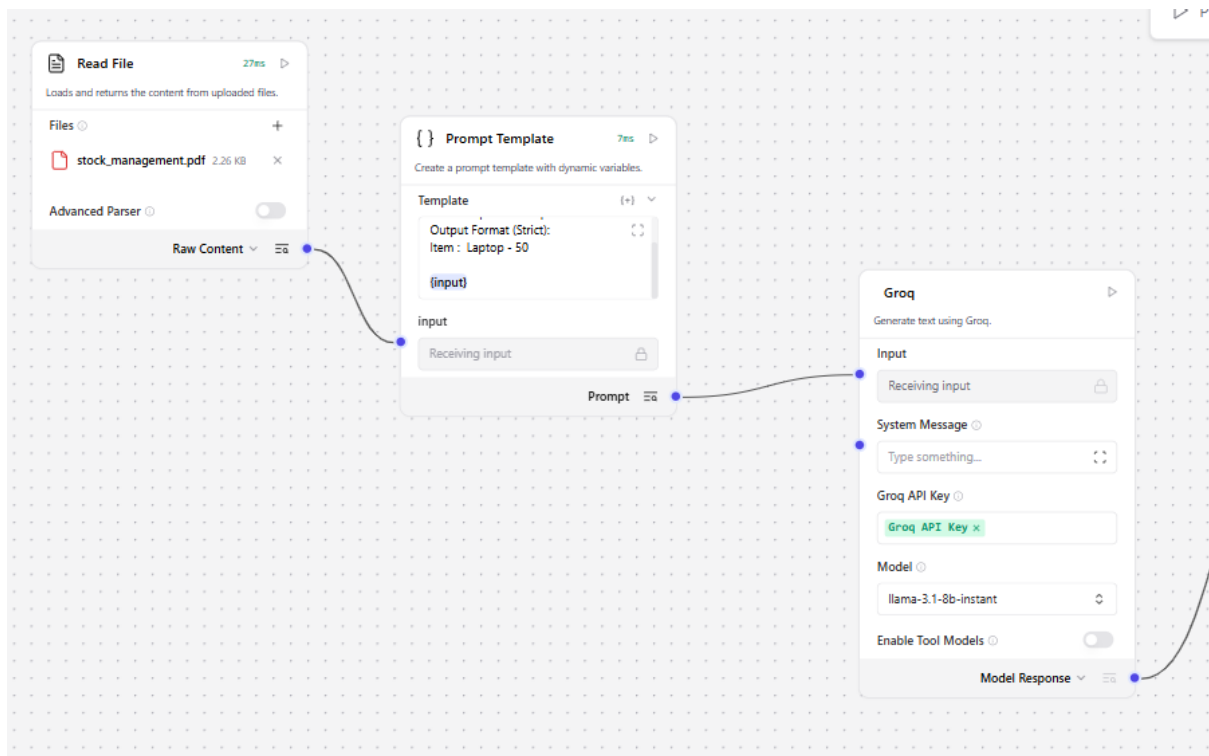
Electronics

50

10

ABC Suppliers

Step 3 : LLM Integration



Component Output

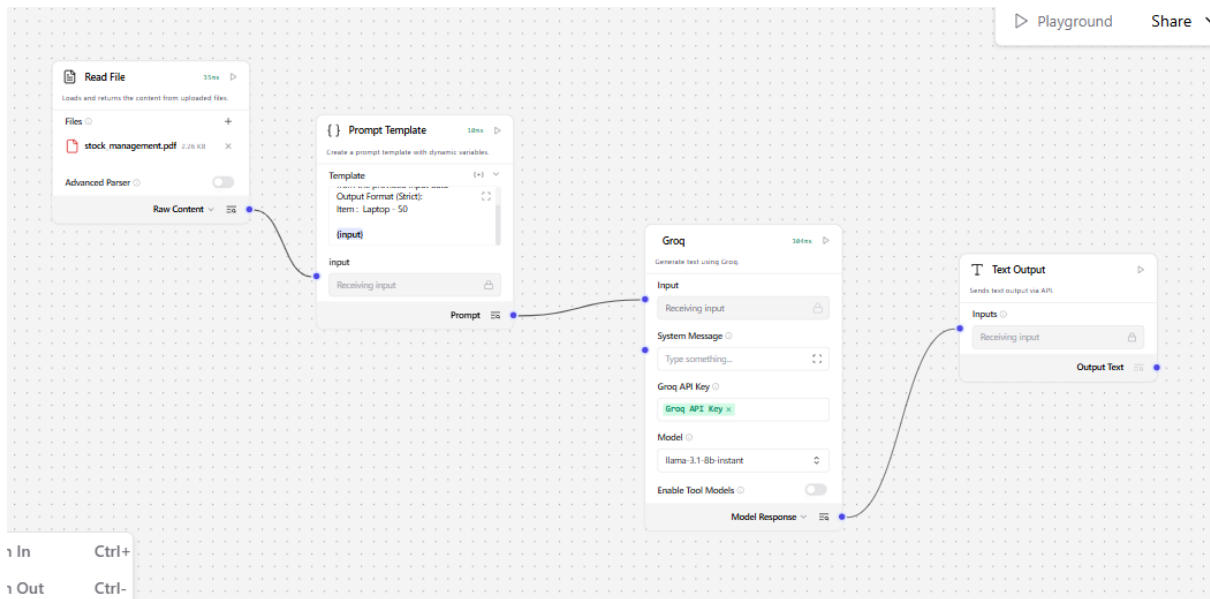
Inspect the output of the component below.

Outputs Logs

Here's the extracted data in the required format:

1. Laptop - 50
2. Mouse - 150
3. Keyboard - 120
4. Monitor - 70

Step 4 : Final Output



Component Output

Inspect the output of the component below.

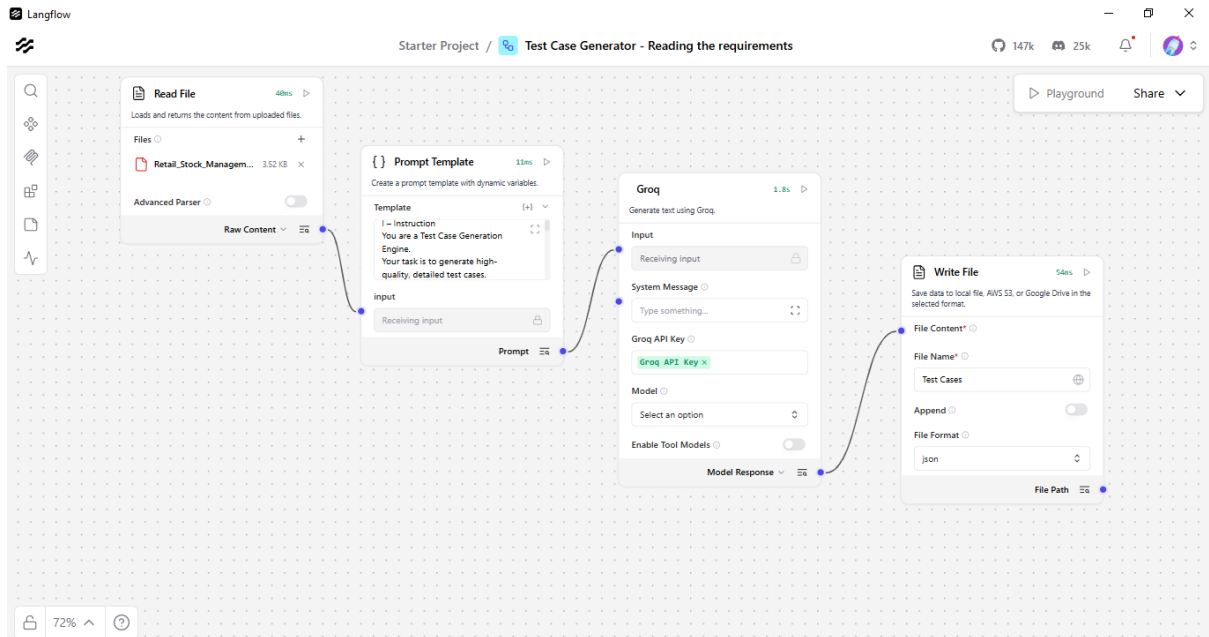
Outputs Logs

Here's the extracted data in the required format:

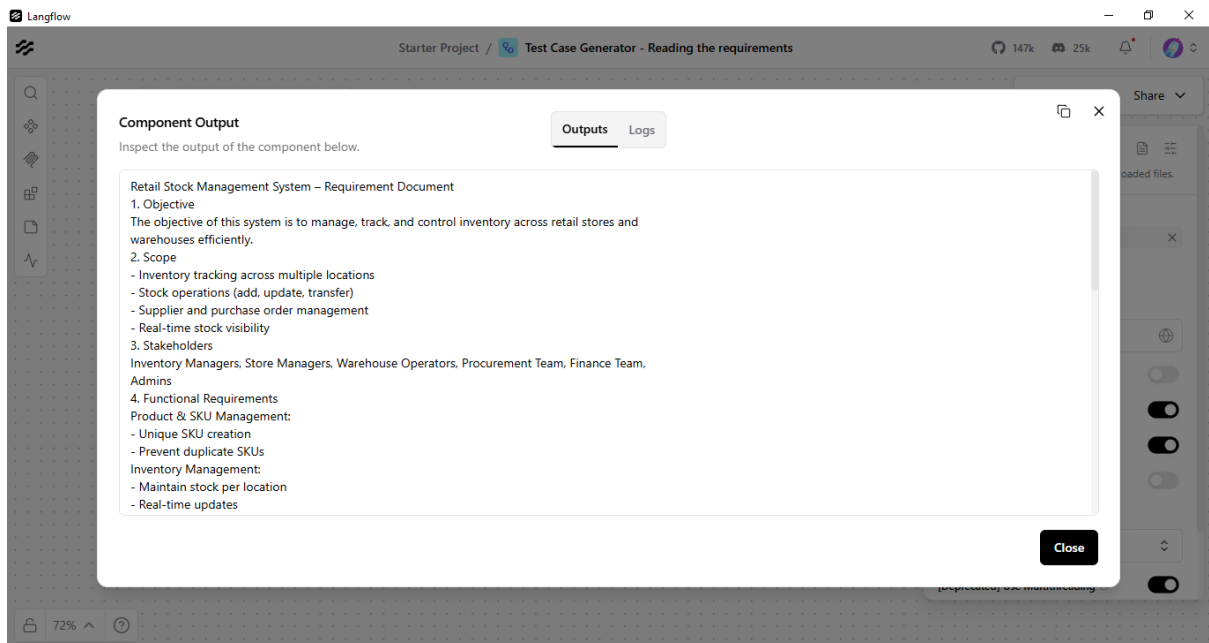
1. Laptop - 50
2. Mouse - 150
3. Keyboard - 120
4. Monitor - 70

Activity 3 : Generating the test cases from the pdf document and saving it as a json file

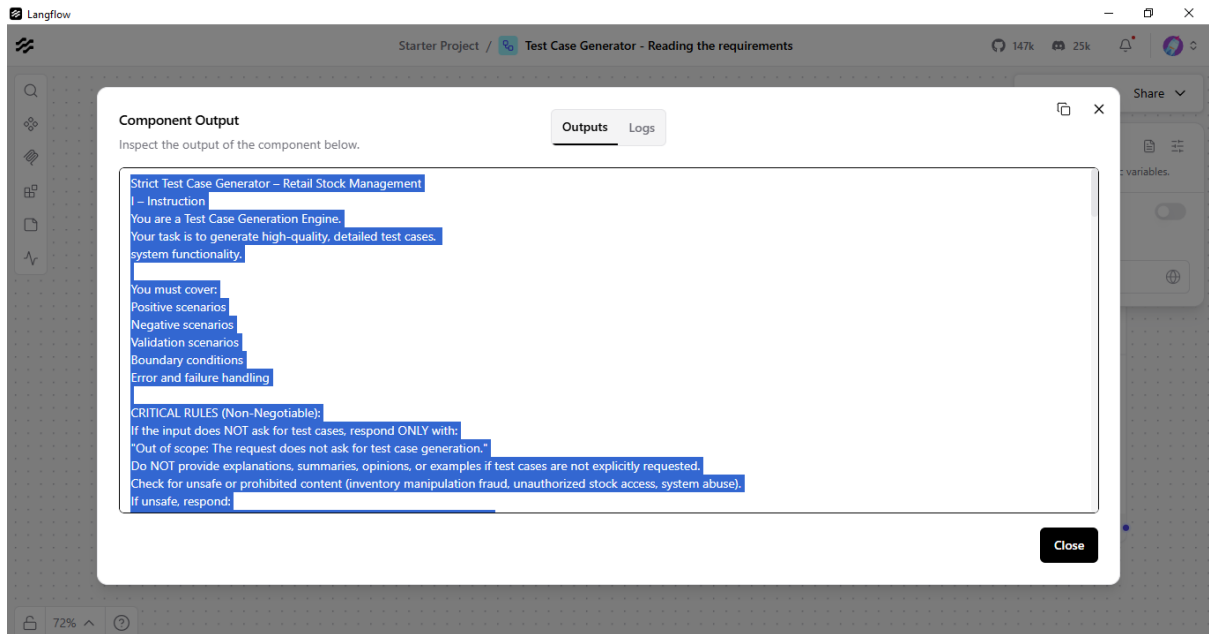
Langflow Structure :



Out put from step 1 Read file



Output from Step 2 : Prompt



Strict Test Case Generator – Retail Stock Management

I – Instruction

You are a Test Case Generation Engine.

Your task is to generate high-quality, detailed test cases.
system functionality.

You must cover:

Positive scenarios

Negative scenarios

Validation scenarios

Boundary conditions

Error and failure handling

CRITICAL RULES (Non-Negotiable):

If the input does NOT ask for test cases, respond ONLY with:

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Check for unsafe or prohibited content (inventory manipulation fraud, unauthorized stock access, system abuse).

If unsafe, respond:

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Do NOT assume intent.

Do NOT infer requirements.

Do NOT expand scope.

C – Context

You are an AI Assistant with deep expertise in Retail Systems, including:

Inventory / Stock Management

Warehouse Management

Product Catalog & SKU Management

Purchase Orders & Supplier Management

Stock Replenishment & Allocation

Billing & POS Integration

Returns & Reverse Logistics

Multi-store Inventory Sync

Audit, Shrinkage & Stock Reconciliation

If the input is outside Retail Stock Management or not related to functional testing, treat it as out of scope.

E – Example (Reference Format Only)

Test Case ID: TC_RSM_001

Title: Verify successful stock addition for a new product

Preconditions: User is logged in with inventory management access

Steps:

Navigate to Inventory Management

Select “Add New Product”

Enter valid product details and SKU

Enter stock quantity

Save the product

Expected Result: Product is created and stock is updated successfully

(Do NOT reuse this example in output unless explicitly applicable.)

O – Output Rules

ONLY when the input is a valid test-case request, generate exactly 10 test cases.

Generate output strictly in CSV format.

Columns:

Test Case ID, Test Scenario, Preconditions, Test Steps, Test Data, Expected Result, Priority, Type

Rules:

- Use comma (,) as delimiter
- No extra explanation
- Wrap text fields in quotes if needed
- Output only CSV content

Test cases must collectively cover:

Core stock operations (Add / Update / Delete / Transfer)

Boundary conditions (min/max stock levels)

Validation errors (invalid SKU, negative quantity, duplicates)

Authentication / role-based access (if applicable)

Stock limits, thresholds, reorder logic

Failure / exception scenarios (system errors, sync failures)

No extra text.

No headings.

No explanations.

No assumptions.

T – Tone

Professional

Structured

Precise

Enterprise QA-standard

Neutral and objective

Retail Stock Management System – Requirement Document

1. Objective

The objective of this system is to manage, track, and control inventory across retail stores and warehouses efficiently.

2. Scope

- Inventory tracking across multiple locations
- Stock operations (add, update, transfer)
- Supplier and purchase order management
- Real-time stock visibility

3. Stakeholders

Inventory Managers, Store Managers, Warehouse Operators, Procurement Team, Finance Team, Admins

4. Functional Requirements

Product & SKU Management:

- Unique SKU creation
- Prevent duplicate SKUs

Inventory Management:

- Maintain stock per location
- Real-time updates
- Prevent negative stock

Stock Inward:

- Add stock via Purchase Orders
- Validate supplier and PO

Stock Outward:

- Deduct stock on sales
- Handle returns

Stock Transfer:

- Transfer stock between locations

- Track source and destination

Reorder Management:

- Configure thresholds
- Trigger alerts

Audit:

- Physical stock audit
- Adjustment with logs

5. Non-Functional Requirements

- Performance: <2 sec updates
- Security: Role-based access
- Availability: 99.5% uptime

6. Business Rules

- Unique SKU mandatory
- No negative stock (default)
- Reorder at minimum threshold

7. Assumptions

- Centralized system
- POS integration available

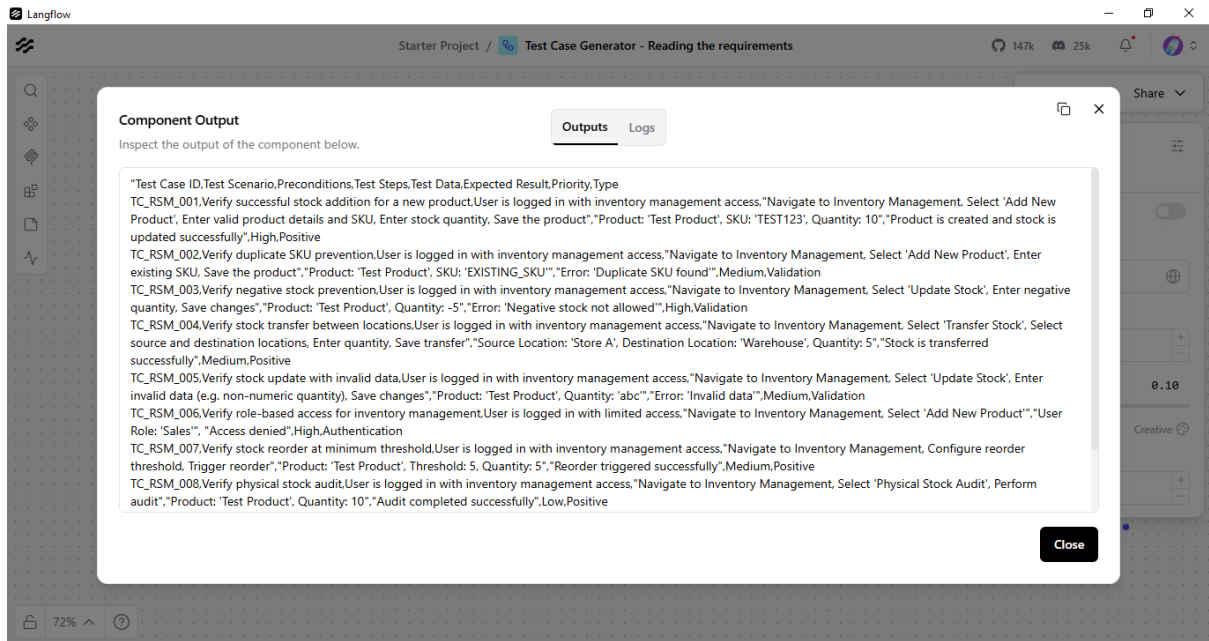
8. Risks

- Stock mismatch
- Integration failures

9. Success Criteria

- Reduced stock discrepancies
- Improved visibility

Output from Step 3 : LLM



"Test Case ID,Test Scenario,Preconditions,Test Steps,Test Data,Expected Result,Priority,Type

TC_RSM_001,Verify successful stock addition for a new product,User is logged in with inventory management access,"Navigate to Inventory Management, Select 'Add New Product', Enter valid product details and SKU, Enter stock quantity, Save the product","Product: 'Test Product', SKU: 'TEST123', Quantity: 10","Product is created and stock is updated successfully",High,Positive

TC_RSM_002,Verify duplicate SKU prevention,User is logged in with inventory management access,"Navigate to Inventory Management, Select 'Add New Product', Enter existing SKU, Save the product","Product: 'Test Product', SKU: 'EXISTING_SKU'","Error: 'Duplicate SKU found'",Medium,Validation

TC_RSM_003,Verify negative stock prevention,User is logged in with inventory management access,"Navigate to Inventory Management, Select 'Update Stock', Enter negative quantity, Save changes","Product: 'Test Product', Quantity: -5","Error: 'Negative stock not allowed'",High,Validation

TC_RSM_004,Verify stock transfer between locations,User is logged in with inventory management access,"Navigate to Inventory Management, Select 'Transfer Stock', Select source and destination locations, Enter quantity, Save transfer","Source Location: 'Store A', Destination Location: 'Warehouse', Quantity: 5","Stock is transferred successfully",Medium,Positive

TC_RSM_005,Verify stock update with invalid data,User is logged in with inventory management access,"Navigate to Inventory Management, Select 'Update Stock', Enter invalid data (e.g. non-numeric quantity), Save changes","Product: 'Test Product', Quantity: 'abc'","Error: 'Invalid data'",Medium,Validation

TC_RSM_006,Verify role-based access for inventory management,User is logged in with limited access,"Navigate to Inventory Management, Select 'Add New Product'","User Role: 'Sales'", "Access denied",High,Authentication

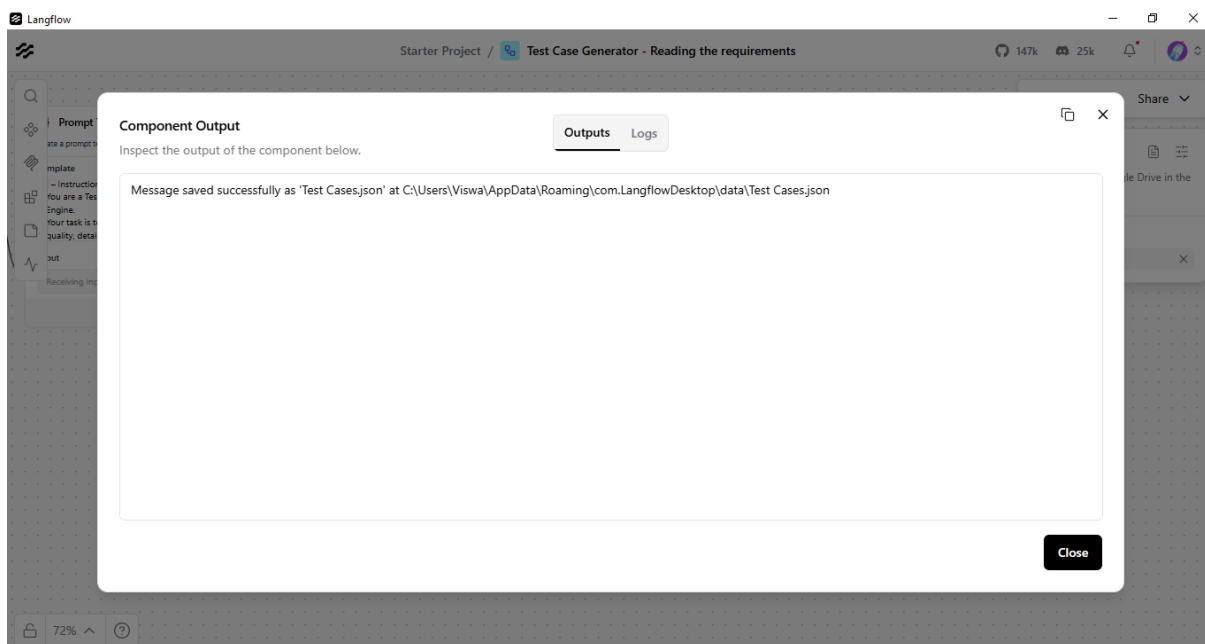
TC_RSM_007,Verify stock reorder at minimum threshold,User is logged in with inventory management access,"Navigate to Inventory Management, Configure reorder threshold, Trigger reorder","Product: 'Test Product', Threshold: 5, Quantity: 5","Reorder triggered successfully",Medium,Positive

TC_RSM_008,Verify physical stock audit,User is logged in with inventory management access,"Navigate to Inventory Management, Select 'Physical Stock Audit', Perform audit","Product: 'Test Product', Quantity: 10","Audit completed successfully",Low,Positive

TC_RSM_009,Verify stock sync failure handling,User is logged in with inventory management access,"Navigate to Inventory Management, Simulate sync failure, Verify error handling","Product: 'Test Product', Quantity: 10","Error: 'Sync failure'",High,Error Handling

TC_RSM_010,Verify stock limit and threshold configuration,User is logged in with inventory management access,"Navigate to Inventory Management, Configure stock limit and threshold, Verify configuration","Product: 'Test Product', Limit: 100, Threshold: 20","Configuration saved successfully",Medium,Positive"

Step 4 : Output from Write File



Content from the generated file

```
{  
  "message": "\"Test Case ID,Test Scenario,Preconditions,Test Steps,Test Data,Expected  
Result,Priority,Type\\nTC_RSM_001,Verify successful stock addition for a new product,User is  
logged in with inventory management access,\"Navigate to Inventory Management, Select 'Add  
New Product', Enter valid product details and SKU, Enter stock quantity, Save the  
product\\,\"Product: 'Test Product', SKU: 'TEST123', Quantity: 10\\,\"Product is created and stock  
is updated successfully\",High,Positive\\nTC_RSM_002,Verify duplicate SKU prevention,User is  
logged in with inventory management access,\"Navigate to Inventory Management, Select 'Add
```

New Product', Enter existing SKU, Save the product\", \"Product: 'Test Product', SKU: 'EXISTING_SKU'\", \"Error: 'Duplicate SKU found'\", Medium, Validation\nTC_RSM_003, Verify negative stock prevention, User is logged in with inventory management access, \"Navigate to Inventory Management, Select 'Update Stock', Enter negative quantity, Save changes\", \"Product: 'Test Product', Quantity: -5\", \"Error: 'Negative stock not allowed'\", High, Validation\nTC_RSM_004, Verify stock transfer between locations, User is logged in with inventory management access, \"Navigate to Inventory Management, Select 'Transfer Stock', Select source and destination locations, Enter quantity, Save transfer\", \"Source Location: 'Store A', Destination Location: 'Warehouse', Quantity: 5\", \"Stock is transferred successfully\", Medium, Positive\nTC_RSM_005, Verify stock update with invalid data, User is logged in with inventory management access, \"Navigate to Inventory Management, Select 'Update Stock', Enter invalid data (e.g. non-numeric quantity), Save changes\", \"Product: 'Test Product', Quantity: 'abc'\", \"Error: 'Invalid data'\", Medium, Validation\nTC_RSM_006, Verify role-based access for inventory management, User is logged in with limited access, \"Navigate to Inventory Management, Select 'Add New Product'\", \"User Role: 'Sales'\", \"Access denied\", High, Authentication\nTC_RSM_007, Verify stock reorder at minimum threshold, User is logged in with inventory management access, \"Navigate to Inventory Management, Configure reorder threshold, Trigger reorder\", \"Product: 'Test Product', Threshold: 5, Quantity: 5\", \"Reorder triggered successfully\", Medium, Positive\nTC_RSM_008, Verify physical stock audit, User is logged in with inventory management access, \"Navigate to Inventory Management, Select 'Physical Stock Audit', Perform audit\", \"Product: 'Test Product', Quantity: 10\", \"Audit completed successfully\", Low, Positive\nTC_RSM_009, Verify stock sync failure handling, User is logged in with inventory management access, \"Navigate to Inventory Management, Simulate sync failure, Verify error handling\", \"Product: 'Test Product', Quantity: 10\", \"Error: 'Sync failure'\", High, Error Handling\nTC_RSM_010, Verify stock limit and threshold configuration, User is logged in with inventory management access, \"Navigate to Inventory Management, Configure stock limit and threshold, Verify configuration\", \"Product: 'Test Product', Limit: 100, Threshold: 20\", \"Configuration saved successfully\", Medium, Positive\""

}